The Fascinating TeV Sky

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The status of very high energy gamma-ray astronomy is described with an emphasis on the astrophysical implications of recent discoveries of galactic sources of gamma-rays in the TeV energy regime. The scientific rationale and potential of the field is briefly discussed in the context of future generation of ground-based detectors.

Keywords: VHE gamma-ray sources, SNRs, PWN, GMCs, Microquasars

1. Introduction

The recent discoveries of tens of galactic and extragalactic sources of very high energy (VHE; $E \geq 100$ GeV) gamma-rats have revolutionized the field – VHE gamma-ray astronomy has emerged as a truly astronomical (observational) discipline[1]. The reports from several (first of all, HESS, MAGIC, MILAGRO and VERITAS) groups using ground-based detectors not only support the earlier theoretical predictions about the great potential of gamma-ray astronomy, but also have revealed many new features concerning, in particular, the unexpectedly high efficiency of particle acceleration in a broad variety of astrophysical environments. The existence of extreme accelerators on different galactic and extragalactic scales, from compact Black Holes and Neutron Stars to large-scale cosmological structures like Galaxy Clusters, makes the fascinating VHE sky an excellent laboratory for the study of complex nonthermal processes in the ultrarelativistic regime.

2. Detection technique

The Earth's atmosphere is not transparent to gamma-rays of any energy. Therefore, their registration requires detectors to be installed on space platforms. However, as long as the latters cannot offer detection areas significantly exceeding 1 m^2, the efficiency of study of tiny fluxes of TeV gamma-rays with space-based instruments are quite limited. Fortunately, at higher energies an alternative method can be used for detection of gamma-rays. The method is based on the registration of atmospheric showers (initiated by interactions of gamma-rays) either directly or through their Cherenkov radiation. The faint and brief Cherenkov flashes which last only several nanoseconds can be detected by large optical reflectors equipped with fast multipixel cameras. With a telescope consisting of an optical reflector of diameter $D \approx 10$ m, and a multichannel camera with a pixel size $0.1^\circ - 0.2^\circ$ and a field-of-view $\Theta \geq 3^\circ$, primary gamma-rays of energy ≥ 0.1 TeV can be collected from distances as large as 100 m. This provides huge detection areas, $A \geq 10^4$ m^2, which largely compensate the small TeV gamma-ray fluxes. The total number of photons in the registered Cherenkov light image is a measure of energy, while the orientation of the image correlates with the arrival direction of the gamma-ray. Finally, the shape of the image contains information about the origin of primary particle (a proton or photon). The stereoscopic observations of air showers[2] with two or more telescopes located at distances of about 100 m from each other provide effective rejection of hadronic showers (by a factor of 100), very good (better than 0.1°) angular resolution and reasonable (better than 15 per cent) energy resolution. At energies around 1 TeV, this results in a minimum detectable energy flux as low as 3×10^{-13} erg/cm^2. The sensitivity of IACT arrays is much better than in any other gamma-ray domain, including the GeV energy band, where the sensitivity of *Fermi* Large Area Telescope (hereafter Fermi LAT)[3], even after dramatic improvement compared to the performance of the previous gamma-ray space-borne instruments, still cannot compete with the performance already achieved in the TeV energy band. Thanks to very large collection area, the IACT technique provides adequate gamma-ray photon statistics. Coupled with good energy and angular resolutions, the rich gamma-ray photon statistics allows deep morphological, spectral, and temporal studies. This makes the IACT arrays perfect multifunctional and multi-purpose astronomical tools for exploration of a broad range of nonthermal phenomena.

The IACT arrays are designed for observations of point-like or moderately extended (with angular size 1° or less) objects with known celestial coordinates. However, the high sensitivity and relatively large ($\geq 4°$) field of view of IACT arrays allow effective all-sky surveys. In particular, a large number of TeV galactic sources (including the ones not yet unidentified) with energy fluxes as low as 10^{-12} erg/cm^{2}s have been discovered in the HESS survey of the galactic plane. On the other hand, the potential of IACT arrays is relatively limited for the search for very extended structures, like diffuse emission, associated, for example, with the inner Galaxy. The capability of IACT arrays is limited also for the searches for solitary events or highly variable phenomena in the very-high-energy regime. In this regard, the ground-based gamma-ray detection technique based on direct registration of particles that comprise the extensive air showers (EAS), is a complementary approach to the IACT technique. The tradiational EAS technique, based on particle detectors, e.g. scintillators, spread over large areas, works effectively for detection of cosmic rays at ultra-high energies, $E \geq 100$ TeV. In order to make this technique more relevant to the purposes of gamma-ray astronomy, the detection energy threshold should be reduced at least by an order of magnitude. This can be achieved using dense particle arrays or large water Cherenkov detectors located on very high altitudes. The feasibility of both approaches recently have been successfully demonstrated by the Tibet AS$_\gamma$ and Argo collaborations, and especially by the Milagro group which has reported statistically significant detections of diffuse multi-TeV gamma-ray emission from different parts of the galactic plane[4].

3. TeV gamma-ray source populations

The rapidly increasing number of sources in the TeV sky is an objective indicative of the progress of the field. However, the most striking feature of VHE gamma-ray astronomy is not the number of sources (which remains still quite modest, around 100), but rather the number of source populations (classes) represented by the reported TeV gamma-ray emitters. Quite surprisingly, almost all, with a few exceptions, astrophysical source populations, which have been established or suspected to be sites of acceleration of relativistic particles, appeared to contain objects characterized by emission effectively extending to TeV energy domain. Moreover, in some cases the VHE gamma-ray domain dominates over the other energy bands of these objects implying that we deal with extreme nonthermal machines in which particles are boosted to VHE energies with an almost 100 % efficiency. In our Galaxy, at least four well-established source classes contain TeV gamma-ray emitters: Supernova Remnants, (SNRs), Pulsar Wind Nebulae (PWNe), Binary Pulsars (BPs), Giant Molecular Clouds (GMCs). Note that most of the TeV galactic sources appear to be extended, from an angular size of several arcminutes to several degrees. This makes their identification quite difficult, and explains, to a certain extent, the the large fraction of unidentified TeV gamma-ray sources. While some of these objects might have direct or indirect links to the above mentioned source populations, one cannot exclude that some of the unidentified TeV sources represent a new, yet unknown, class of astrophysical objects. In this regard, the exploration of the origin of these sources is a great interest in its own right.

The Identification of compact extragalactic sources is a simpler task, given the small angular size, as well as the characteristic variability of radiation which generally correlates with the light curves obtained at other wavelengths. So far, the majority of reported extragalactic sources are the so-called blazars - Active Galactic Nuclei (AGN) with relativistic jets directed towards the observer. In addition, the catalog of extragalactic sources contains at least two representatives of radio galaxies - M87 and Centaurus A, and two representatives of Starburst Galaxies - M82 and NGC 253. In this paper I focus on the Galactic Sources of TeV gamma-ray emission. The status of the field regarding the Extragalactic Sources and related topics like the Extragalactic Background Radiation are discussed in the rapporteur paper by B. Giebels, F. Aharonian and H. Sol prepared for the Session "Active Galactic Nuclei and Gamma Rays" of the 12th MG meeting[5].

4. SNRs and Origin of Galactic Cosmic Rays

It is generally believed that gamma-ray astronomy has a key role to play in solving the problem of origin of galactic cosmic rays. The basic concept formulated by pioneers of the field in the 1950's and 1960's is simple and concerns both the acceleration and propagation aspects of cosmic rays. One of the key objectives in this regard is the test of the hypothesis that supernova remnants (SNRs) are responsible for the major fraction of the observed flux of galactic cosmic rays at energies below the so-called *knee* around 10^{15} eV. The main *phenomenological* argument in favor of this hypothesis is based on the fact that the production rate of galactic cosmic rays $\dot{W}_{\rm CR} \simeq 10^{41}$ erg/s can be be supported by SNRs if a quite reasonable fraction, 10 per cent or so, of the total mechanical energy of SN explosions in our Galaxy is eventually released in cosmic rays[6]. The second key argument in favor of SNRs as sources of galactic CRs has a more theoretical background, namely it relies on the diffusive shock acceleration (DSA) paradigm applied to young SNRs (for a recent review, see Ref. [7]).

Three factors, (i) the high efficiency of transformation of the available kinetic energy of bulk motion into nonthermal particles, (ii) the hard energy spectra of protons extending to multi-TeV energy region, and (iii) the relatively high density of the ambient gas, make the young supernova remnants viable sources of gamma-rays resulting from the production and prompt decay of secondary π^0-mesons. Thus, the most straightforward test of acceleration of proton and nuclei in SNRs can be performed via search for π^0-decay gamma-rays - either directly from shells of young SNRs or from nearby clouds interacting with an expanding SNR shell. It has been argued[8] that the TeV gamma-ray domain is the best energy band to explore this possibility - from the point of view of the superior performance of the detection technique and because of the key information about particle acceleration carried by TeV gamma-rays.

These theoretical predictions initiated significant efforts towards detection of TeV gamma-rays from SNRs. Presently five young shell type SNRS - Cas A, RX J1713.7-3946, RX J0852-4622 (Vela Jr), RCW 86, and SN 1006 - are unambiguously identified as TeV gamma-ray emitters. Several others, in particular IC 433, W28 and CTB 37B are, most likely, SNR/molecular-cloud interacting systems. Also, several galactic TeV gamma-ray sources spatially coincide with the so-called composite supernova remnants, objects with characteristic features of both standard shell-type SNRs and PWNe (plerions). At least in one case, the association of a TeV source with the composite SNR G0.9+0.1 is clearly established (see e.g. recent review articles Ref.[9,10]).

In Fig.1 is shown the gamma-ray image of the brightest TeV SNR - RX J1713.7-3946 obtained by the HESS telescope array. The overall shell structure and its correlation with the nonthermal X-ray image is clearly visible. It should be noted that all reported TeV gamma-ray emitting SNRs are characterized by synchrotron X-ray emission. In particular, the extension of the spectrum of RX J1713.7-3946 to 10 keV and beyond is an indicator that the diffusive shock acceleration of electrons in this source proceeds at the maximum possible acceleration rate. The large shock speed, $v \sim 3000$ km/s, the high turbulence allowing particles to diffuse in the Bohm regime, are two necessary conditions for acceleration of particles, electrons and/or protons, to ≥ 100 TeV, and correspondingly for extension of gamma-rays beyond 10 TeV. The energy spectrum of gamma-rays integrated over the entire remnant is shown in Fig.2. It extends over two decades in energy, from 300 GeV to ≥ 30 TeV. Perhaps the most principal issue related to the interpretation of radiation of this

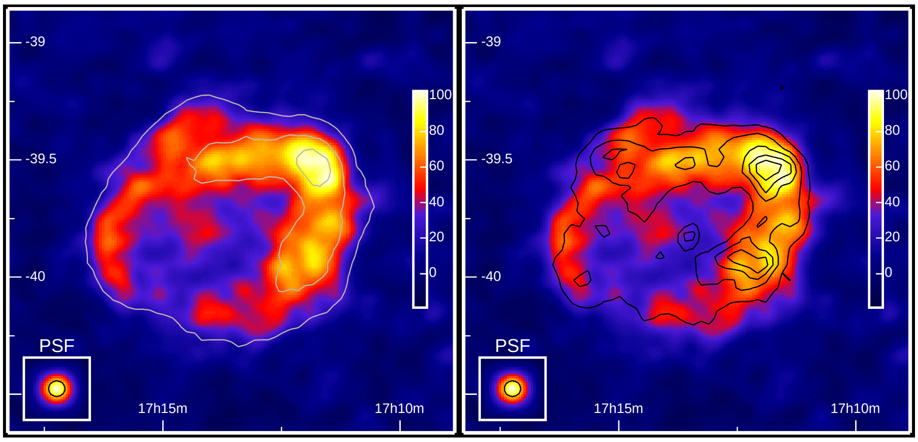

Fig. 1. The TeV gamma-ray image of RX J1713.7-3946 smoothed with a Gaussian kernel of 2 arcmin[11]. On the *right-hand side*, the ASCA 1-3 keV X-ray contours (black lines) are shown for comparison.

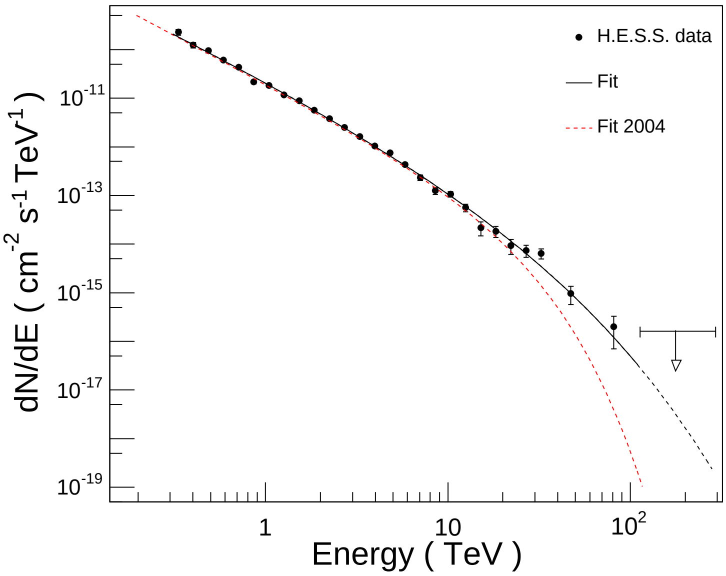

Fig. 2. The energy spectrum of gamma-rays from RX J1713.7−3946. The spectral points are nicely described by a power law with an exponential cutoff of the form $\mathrm{d}N/\mathrm{d}E = I_0\, E^{-\Gamma}\, \exp\left(-(E/E_{\mathrm{c}})^{\beta}\right)$. The best fit (black solid line) is given here for $\beta = 0.5$ (fixed), $\Gamma = 1.8$, $E_{\mathrm{c}} = 3.7$ TeV, and $I_0 = 3.4 \times 10^{-11}$ ph/cm$^{\mathrm{s}}$TeV[11].

source, in particular in the context of ability of SNRs to explain the galactic cosmic rays, is the origin of the TeV gamma-ray emission. Presently both possibilities related to the proton-proton interactions or to the inverse Compton scattering of electrons are considered as viable options. The main hope to discard one of these possibilities is related to detailed studies of the spectral and spatial distributions of the source at other wavelengths, especially at X-rays.

The deep observations of this source with sub-arcsecond resolution of the Chandra X-ray telescope resulted in the discovery of a complex network of bright X-ray filaments and knots, as well as X-ray variability of compact regions on timescales less than a few years[12]. A likely explanation of these features is the significant, by a factor of 10 or more, enhancement of the magnetic field compared to the interstellar magnetic field, e.g. through the streaming instability[13].

In a simple one-zone model the large magnetic field ($\geq 10\mu$G) gives the preference to the hadronic origin of gamma-rays, given that the electrons are cooled basically through the synchrotron radiation, and thus only a negligible fraction of the electron energy is released via inverse Compton scattering. However, in more sophisticated models, e.g. with reverse shocks[14], the TeV gamma-ray emission can be explained by multi-TeV electrons. The future spectral measurements at GeV and multi-TeV energies (above 10 TeV) are crucial for the conclusion whether the gamma-rays are produced by accelerated protons or electrons. In fact, the picture could be much more complex, e.g. with significant contributions from both forward and reverse shocks. The hadronic radiation also might have quite clumpy character contributed mainly from from dense gas regions in the shell[14].

Within the *diffusive shock acceleration* paradigm, the shock speed exceeding a few 1000 km/s is a key condition for acceleration of particles to multi-TeV energies. Since this condition is satisfied only in SNRs with age of ≤ 3000 yr, the young SNRs are prime candidates for TeV gamma-ray emission. On the other hand we may expect gamma-rays from older SNRs - due to interactions of highest energy protons that already left the remnant with the nearby dense gas environments like Giant Molecular Clouds (GRBs)[15] or from shocked clouds overrun by a blastwave[16] Two nice example of such sources are the middle-age SNRs W28 and IC 433. Both source have been detected in TeV gamma-rays by the HESS, MAGIC and VERITAS collaborations, and recently also in MeV/GeV gamma-rays by the *AGILE* and *Fermi* groups. These objects have complex morphologies (see Fig.3) related, most likely, to the character of propagation of cosmic rays after they left the remnant, or to the interaction of the cloud and the supernova blastwave. In particular, while the extended GeV source within the boundary of SNR W28 found by the AGILE[17] and Fermi LAT groups[18] extensively coincides with the TeV gamma-ray source HESS J1801-233[19], one can see local variations of the GeV to TeV flux ratio outside the SNR. This implies a difference of the CR spectra in the north-west and south molecular cloud complexes[17].

5. Pulsar Wind Nebulae

Pulsars are sites of effective transformation of rotational energy of highly magnetized neutron stars into electromagnetic waves (Poynting flux), and eventually to relativistic particles. The complex nonthermal processes related to pulsars occur in

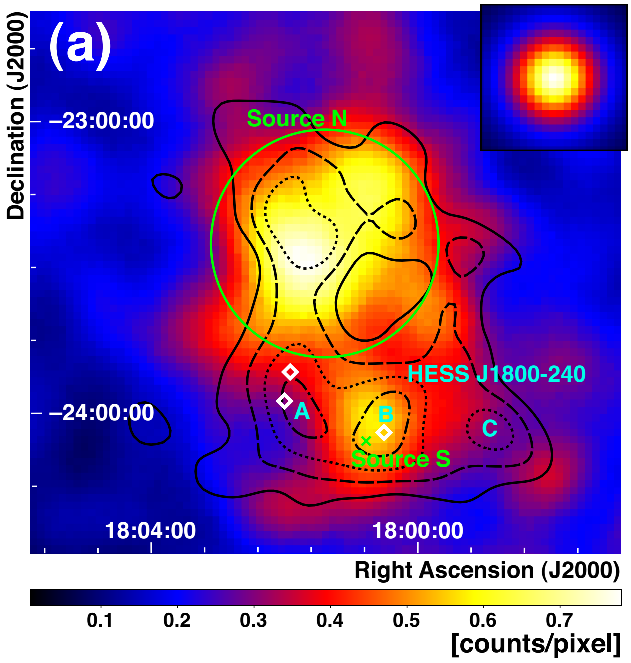

Fig. 3. The smoothed image of 2-10 GeV gamma-rays around W28 obtained with the Fermi LAT[18]. Black contours show the HESS significance map for TeV gamma-rays (20%, 40%,60% and 80% of the peak value). The bright HESS (TeV) spots A,B and C[19] are also indicated.

three physically distinct sites: (i) compact regions of *Pulsar Magnetosphere* where magnetic field can be as large as 10^{12} G, (ii) *Unshocked Ultrarelativistic Wind* which carries almost the entire rotational energy of the pulsar in the form of Poynting flux and kinetic energy of the bulk motion, and (iii) *Pulsar Wind Nebula* – the result of termination of the MHD wind with subsequent acceleration and radiation of relativistic electrons.

The Crab Nebula is the most prominent, although not an archetypical (as it is often claimed in the literature) representative of PWNe. It is, in fact, a unique object with a spectrum of nonthermal radiation extending from radio wavelengths to very high energy gamma-rays. The currently accepted standard MHD wind model of the Crab Nebula assumes that the pulsar wind is terminated by a standing reverse shock which accelerates electrons to $\geq 10^{15}$ eV and randomizes their pitch angles. The major fraction of the rotational energy of the pulsar is eventually released in nonthermal synchrotron radiation extending to the hard X-ray/low-energy gamma-ray energy domain, thus around 1 GeV we should expect a transition from the Synchrotron component to the IC component of gamma-rays. This prediction has been recently confirmed by observations of FERMI LAT[20] (see Fig.4). While the synchrotron radiation of $\geq 10^{14}$ eV electrons extends to the multi-MeV region, the inverse Compton scattering of same electrons results in ultra-high energy gamma-radiation, up to $\geq$ 50 TeV as reported by the HEGRA, HESS and CANGAROO collaborations. Remarkably, the fluxes around 100 GeV obtained by the Fermi LAT[20] and MAGIC[21] telescopes based on different detection techniques is in very good agreement and confirm the spectral shape of the IC component predicted theoretically.

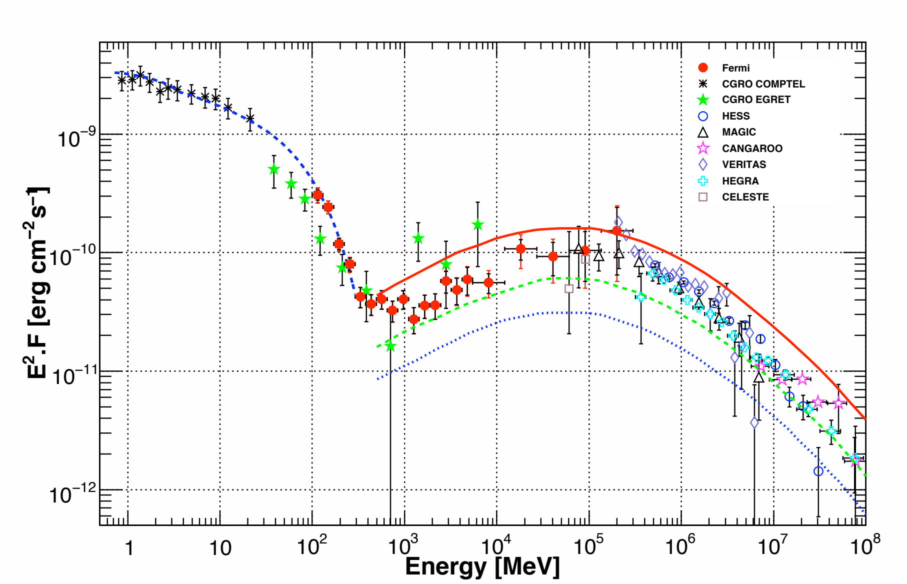

Fig. 4. Broad-band spectral energy distribution (SED) of the Crab Nebula (see for details ref. [20]). The termination shock of the pulsar wind is an extreme accelerator boosting the energy of electrons to 10^{15} eV and beyond. In the framework of the synchrotron/IC emission model, it is possible to explain the entire energy range of radiation, from radio wavelengths to ultra-high energy gamma-rays.

PWNe constitute an well established TeV gamma-ray source population with almost 20 representatives from which more than half are reliably identified with well known objects. The Cosmic Microwave Background Radiation (CMBR) provides universal target photon field in PWNe for production of IC gamma-rays by multi-TeV electrons. On the other hand, the strength of the magnetic field has strong impact on production of gamma-rays, suppressing dramatically the fraction of energy channeled through IC scattering. In particular, in the Crab Nebula with the magnetic field of order of 100μG, less than 1 percent of the energy of accelerated electrons is converted into IC gamma-rays, the rest being emitted in the form of synchrotron photons. Only the huge energy released in relativistic electrons compensates the low efficiency of gamma-ray production, and thus sustains the gamma-ray luminosity on a detectable level. In other PWNe powered by less luminous pulsars, the resulting nebular magnetic fields are typically on one or two orders of magnitude smaller than in the Crab Nebula making these objects more effective gamma-ray emitters. The main target photons for inverse Compton scattering in these objects is contributed by CMBR, thus the radiative losses of electrons are shared between the synchrotron and IC channels in a fair proportion, $L_\gamma/L_{\rm X} = w_{\rm MBR}/w_{\rm B} \simeq 1\,(B/3\mu{\rm G})^{-2}$. This simple argument makes PWNe potential sources of TeV gamma-rays. More specifically, the plerions powered by pulsars with the "spin-down flux" $L_0/4\pi d^2 \geq 10^{33}$ erg/s kpc^2, become visible in TeV gamma-rays unless the ambient magnetic field exceed 10 μG. The discoveries by HESS and recently also by the Milagro collaboration of TeV gamma-rays from regions located close to pulsars with high spin-down luminosities seem to support this prediction. Some of these sources are identified with famous plerions like MSH 15-52 and the nebula associated with the pulsar PSR J1826-1334. The TeV gamma-ray images of these objects are shown in Fig.5.

The TeV plerions have a distinct feature; the extended gamma-ray regions are, as a rule, offset from the pulsar positions. This is an indication of anisotropic spatial distribution of ultrarelativistic electrons presumably caused by propagation of a reverse shock created at the termination of the pulsar wind in highly inhomogeneous medium. Asymmetric, one-sided images of PWNe have been found in X-rays as well, although on smaller scales. There is at least one fundamental reason for a larger extension of TeV sources compared to their X-ray images. The very fact of detectable gamma-ray emission implies weak magnetic fields in these objects, typically $B \sim 3\mu$G or less, as it follows from the observed X-ray to gamma-ray flux ratios. For these magnetic fields, the electrons responsible for synchrotron X-ray emission are more energetic compared to electrons producing TeV gamma-rays of IC origin. Correspondingly, the synchrotron burning of highest energy electrons makes the X-ray production region relatively compact compared to the gamma-ray production region. For the same reason one should expect energy-dependent morphology of PWNe both in X-rays and TeV gamma-rays caused by energy losses and propagation effects of multi-TeV electrons.

Finally, one should mention the interesting possibility of association of some of the ultrahigh energy gamma-ray sources reported by the MILAGRO collaboration with the PWN Boomerang and with an extended region around the Geminga pulsar[4]. If these associations are real, one has to invoke new non-standard ideas to explain the extension of the gamma-ray spectra beyond 30 TeV. In particular, the IC origin of this radiation would require magnetic field significantly weaker than the interstellar magnetic field. Alternatively the hadronic origin of these gamma-rays requires very large amount of ultra-high energy protons, either as relicts from first epochs of explosion of supernovae or protons and nuclei directly related to the pulsar winds. Both options pose challenges to conventional models of pulsars and their winds. This makes the detailed morphological and spectral studies of these objects a "hot topic" which can be best conducted with stereoscopic Cherenkov telescope arrays designed and optimized for the energy domain of gamma-rays above 10 TeV.

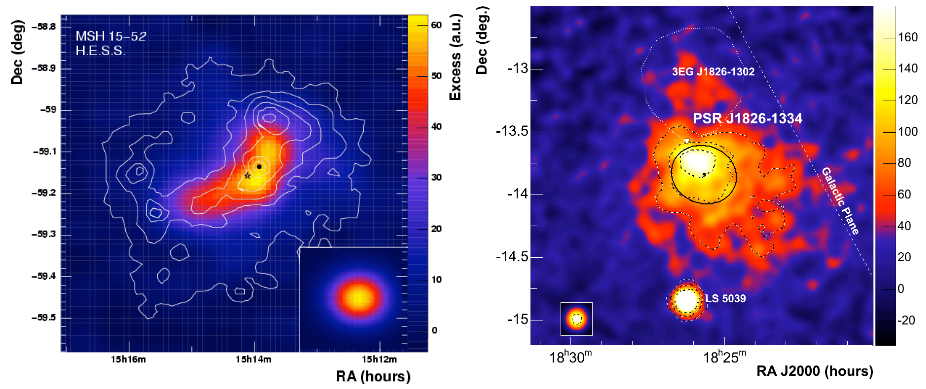

Fig. 5. TeV gamma-ray images of two PWNe detected by the HESS telescope array: MSH 15-52 (left) and the nebula around the pulsar PSR J1826-1334 (right).

6. TeV Binaries

One of the remarkable results of VHE gamma-ray astronomy of recent years is considered the discovery of TeV gamma-rays from binary systems. Despite the controversial history related to claims of detections of gamms-rays from these objects (see e.g. ref.[2]), presently a sub-population of binary systems consisting of a compact object and a very massive star, are established as effective gamma-ray emitters. So far, gamma-rays are detected from three binary systems, and while one of them, PSR B1259-63, is firmly identified as a binary pulsar, the origin of the compact objects in two other systems, LS 5039 and LS I+61 303, remains highly uncertain (for a review see ref.[22]). Two other candidates are Cyg X-1, a classical representative of compact X-ray binaries containing black holes, and an unidentified TeV gamma-ray source HESS J0632+057. Recently, the *Agile collaboration* has claimed an episodic gamma-ray detection from Cyg X-1 at MeV/GeV energies. However one should note that while the reported gamma-ray signals from Cyg X-1 have marginal statistical significance, in the case of HESS J0632+057 the identification of the TeV source with a binary system is quite likely, but not yet firmly established fact.

If a particle accelerator is located in a binary system with a luminous optical star, the interactions of accelerated electrons with the starlight or with the dense stellar wind proceed on time scales of hours or even less. Thus such systems allow "on-line" watch of the complex acceleration and magnetohydrodynamic processes such as the creation and termination of relativistic outflows related to the compact object. This may be a cold pulsar wind in the case of a neutron star or a hot jet in the case of a black hole.

The binary systems containing young, high spin-down luminosity pulsarsn work like a compact PWNe located in high radiation field environment. The best candidate from this class of sources for gamma-ray observations is PSR B1259-63/SS2883 - a binary system consisting of a 48ms pulsar in highly eccentric orbit around a massive B2e companion star. The HESS observations of this source in 2004, from a few days prior the periastron to several months after the periastron passage, revealed a TeV gamma-ray signal with a high statistical significance. The flux was found to vary significantly on timescales of days. Strongest emission signals were observed in pre- and post-periastron phases with indication of flux decrease towards the periastron (unfortunately the source was not observed, due to the moon and bad weather, during the periastron passage). The reported time-averaged energy spectrum of the source is quite steep with a power law photon index ~ 2.7. The detection of TeV γ-rays provides unambiguous evidence for particle acceleration to multi-TeV energies in the binary system, although the origin of gamma-rays remains uncertain. The inverse Compton scattering seems to be the most plausible gamma-ray production mechanism, but neither the minimum around nor the high flux states before and after the periastron agree with the theoretical predictions. To achieve a satisfactory agreement one should introduce additional nonradiative (e.g. adiabatic) losses which dominate over the entire trajectory of the pulsar with significant increase towards the periastron, or assume "early" (sub-TeV) cutoffs in the energy spectra of electrons due to the enhanced rate of Compton losses close to the periastron. An alternative explanation of the observed TeV lightcurve is also possible assuming that gamma-rays are produced at hadronic interactions when the pulsar crosses the stellar disk.

The photon-photon pair production of gamma-rays in the field of optical radiation may play an important role in the formation of gamma-ray lightcurves in the binary systems containing luminous companion stars. Although in the case of PSR 1259-63 the absorption effect appears not significant, in two other binary systems reported as TeV gamma-ray emitters - LS 5039 and LS I+61 303 the strong impact of the photon-photon pair production is unavoidable. Because of the orbital variation of the optical depth, this mechanism should lead to modulation of the TeV gamma-ray flux. The second effect related to the target photon field is the anisotropic IC scattering which makes the predicted TeV lightcurve closer to the observed one. Remarkably, periodic components have been found in the TeV radiation of both sources, in particular from LS 5039 which behaves as a perfect "TeV clock"; the reported gamma-ray period of 3.908d coincides with the well known orbital period of the system. Later, the observations of the Suzaku satellite revealed significant modulation of the X-ray flux of LS 5039, and again, with the same period (see Fig.6). Unlike gamma-rays, the periodic character of the X-ray signal cannot be explained by the effects related to the target photon field. The modulation of synchrotron X-rays can be related only to the periodic variation of the magnetic field and/or the number of relativistic electrons caused, for example, by adiabatic energy losses. The combination of these three effects allows a reasonable interpretation of of lightcurves of both TeV gamma-rays and X-rays, as well as explains the general spectral characteristics in both energy bands[23]. However, many details, in particular the origin of the GeV gamma-ray emission reported recently by the Fermi LAT[24] remains unknown and highly confusing.

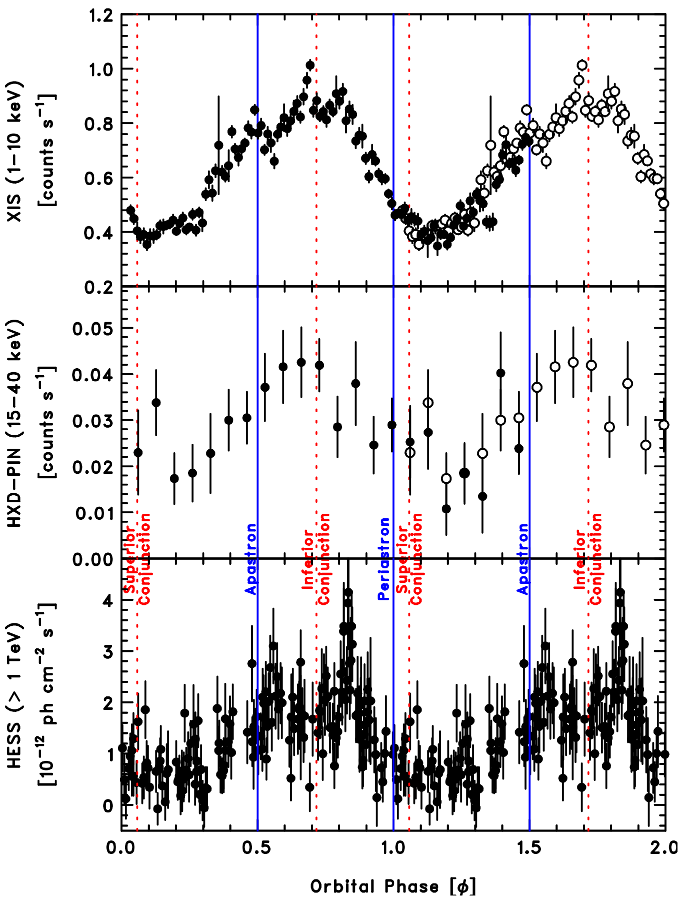

Fig. 6. X-ray and TeV gamma-ray orbital lightcurves of LS 5039. *Suzaku* count rates in the 1–10 keV (top) and 15-40 keV (middle) bands continuously obtained for a 200 ks duration (filled circles). The light curve of integral fluxes at energies E> 1 TeV (bottom) is obtained by HESS on a run-by-run basis.

7. Future Perspectives

The developments of the ground-based detection technique are expected to proceed in three complementary directions, namely towards (*i*) improvement of the sensitivity around 1 TeV by an order of magnitude, (*ii*) reduction of the angular resolution to 1 to 2 arcminutes, and (*iii*) extension of the energy coverage, down to 10 GeV and up to 100 TeV. If one limits the energy threshold to 100 GeV, the performance of the telescope arrays can be predicted with confidence. Namely, a flux sensitivity well below 10^{-13} erg/cm^{2}s can be achieved by a stereoscopic array consisting of tens of 10m-diameter class IACTs. On the other hand, the future telescope arrays would gain a lot if the energy threshold can be reduced to 10-30 GeV. This can be achieved by somewhat larger 15m to 20m diameter class telescopes installed at high mountain altitudes. One may predict, based on the extrapolation of the performance of the current telescope arrays, that two major future projects under consideration should be able to discover and resolve hundreds, or perhaps even thousands of galactic TeV sources.

Complementary to the current attempts towards the realization of large scale stereoscopic arrays of atmospheric imaging Cherenkov telescopes, like CTA and AGIS, the prospects of continuous monitoring of a significant part of the sky, which might lead to exciting discoveries of yet unknown VHE transient phenomena in the Universe, justifies the new proposals of construction of a large field-of-view high altitude EAS detectors like *HAWK*. The expected sensitivity of *HAWK* in the energy region around 1 TeV is expected to be comparable to the sensitivity of Fermi LAT around 1 GeV. In this regard *HAWK* will be complementary to Fermi LAT for continuous monitoring of more than 1 steradian fraction of the sky over three decades of energy from 100 GeV to 100 TeV.

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